



MACHINE LEARNING BASED BUYER SEGMENTATION AND INVESTMENT PROFILING FOR REAL ESTATE MARKET INTELLIGENCE

FELLOWSHIP PROJECT REPORT

Domain:

Financial Analytics & Real Estate Market Intelligence

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Abstract

This study is meant to construct a machine learning model that will be used to classify buyers and their respective investments in the real estate market. Property transaction data, together with other relevant data related to demographics of potential clients, has been analysed in order to classify buyers into different categories depending on their purchasing habits, investment preferences, financing options, and type of properties they choose. Preprocessing of this data was done through encoding, cleaning, and scaling before application of the K-Means Clustering Algorithm to the dataset. The appropriate number of clusters was estimated using the Elbow Method. Evaluation of the clusters was achieved by computing the Silhouette Score. In total four clusters were established, including investment-focused, budget buyers, premium residential buyers, and luxury buyers. A Streamlit interface was created to showcase cluster distribution, investment styles, geographical buyers, and cluster-specific information.

Introduction

Real estate is one of the most economically impactful industries because of its contribution to economic development, investment, and money-making. With the ongoing development in this industry, an understanding of buyer behaviour became crucial not only for companies and firms but for investors and analysts who want to explore the market thoroughly. Buyers might vary in many aspects including purchasing preferences, investment goals, finance ability, and others. Therefore, it might be difficult for a company to build an appropriate marketing strategy or service.

As a rule, traditional methods of gathering demographic information have been used by real estate businesses before. However, the increased amount of data regarding transactions and customers' information makes old approaches less effective because they cannot identify complicated patterns and relations among variables. That is why the use of data analytics in real estate businesses is quite relevant today.

Machine Learning (ML) has become an effective means of processing vast amounts of data and finding important patterns in it. There exist many ways of using Machine Learning, but one of the popular methods in terms of business analytics includes the use of clustering algorithms that allow the classification of people having similar characteristics irrespective of labels assigned before hand. In turn, segmentation of buyers into several groups will help companies better understand consumer preferences, increase marketing efficiency, and manage resources more effectively.

The topic of this project – "Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence" implies the use of Machine Learning methods for analysis of data about buyers of the real estate market and segmentation of customers. For this purpose, such information as consumer demographics, motives, financial preferences,

properties of interest and other factors affecting consumers' decisions is used. As for the classification algorithm, the K-Means method is applied.

Apart from using clustering analysis in this research project, another feature used is that of visualization, which is incorporated using the interactive Streamlit Dashboard. The features included in the dashboard give information on buyer segmentation, investments, geographical distributions, and clustering statistics. With this, it becomes easier to analyze trends in the market through visualization and interactive analytics using machine learning tools.

Some of the results obtained from this study can be very helpful to real estate companies, investors, and market researchers in analyzing customer segments, developing marketing strategies, analyzing investment potential, and making decisions based on insights. This project shows how data analytics and machine learning can be applied in the real estate industry.

Problem Statement

Parcl Co. Limited works in the highly competitive real estate business where knowledge about customer behaviour is an essential tool in decision-making and development of marketing strategies. At the moment, however, Parcl Co. Limited does not have an adequate information system to analyse customer behaviour and provide valuable recommendations. This means that the company does not benefit from many useful insights that could help improve its operations in the future.

Namely, the company needs to identify:

- Types of real estate customers and their buying behaviour.
- Reasons why particular customers make investments.
- Regional patterns of investments in real estate.
- Financing preferences of customers, such as the use of loans.

Failure to identify important characteristics of customers leads to numerous business problems, among which the following should be noted:

- Non-optimal allocation of marketing budget.
- Non-personalised advice about property purchase.
- Difficulties in targeting profitable customers.
- Inadequate personalisation of marketing efforts.
- Lack of understanding of current trends.

A machine learning technique can help solve these problems by clustering customers into distinct categories and analysing their investment behaviour patterns. With this approach, the company will benefit from interactive market intelligence.

Objectives of the Study

- To study demographic details of buyers and sales records of properties.
- To differentiate among various buyers' segments based on machine learning algorithms.
- To comprehend the investing behaviour and financing practices of different customer segments.
- To compare regional differences in terms of real estate investing practices.
- To develop customer segments to assist marketing efforts and investors.
- To design a dashboard to visualise the results of buyer segmentation analysis.

Dataset Description

The data set used in this research was obtained from the case study on the real estate market intelligence. There are two main sets of data involved in this study; namely Client Dataset and Property Dataset. The two data sets have been merged based on a common client identifier to form an integrated data set.

1. Client Dataset

The Client Dataset contains demographic and behavioural information about property buyers. Key attributes include:

- Client Reference ID
- Client Type
- Country
- Gender
- Acquisition Purpose (Home/Investment)
- Loan Applied Status

This dataset helps in understanding buyer characteristics, investment motivations, and financing preferences.

2. Property Dataset

The Property Dataset contains information related to property transactions and property characteristics. Key attributes include:

- Listing ID
- Tower Number
- Unit Category
- Unit Number
- Floor Area (sq. ft.)
- Sale Price
- Listing Status
- Transaction Date

This dataset provides insights into property values, transaction patterns, and buyer purchasing behaviour.

Final Dataset

After merging and preprocessing, the final dataset contained both buyer and property-related attributes, enabling a comprehensive analysis of customer segments. The dataset was further cleaned, encoded, and standardised before applying the K-Means clustering algorithm for buyer segmentation.

The key variables used for clustering included:

- Client Type
- Country
- Gender
- Acquisition Purpose
- Loan Applied
- Floor Area (sq. ft.)
- Sale Price

Methodology

Step 1 – Data Cleaning

- Merged client and property datasets.
- Checked for missing values and inconsistencies.
- Removed duplicate records.
- Standardised data formats.

Step 2 – Feature Encoding

- Applied Label Encoding to categorical variables:
 - Client Type
 - Country
 - Gender
 - Acquisition Purpose
 - Loan Applied Status

Step 3 – Feature Scaling

- Applied StandardScaler to normalise numerical variables such as:
 - Sale Price
 - Floor Area (sq. ft.)

Step 4 – Clustering Model Selection

- Implemented K-Means Clustering for buyer segmentation.
- Chosen due to its efficiency and interpretability for customer segmentation tasks.

Step 5 – Optimal Cluster Selection

- Used the Elbow Method to determine the optimal number of clusters.
- Evaluated cluster quality using the Silhouette Score.
- Selected K = 4 as the optimal number of clusters.

Step 6 – Cluster Interpretation

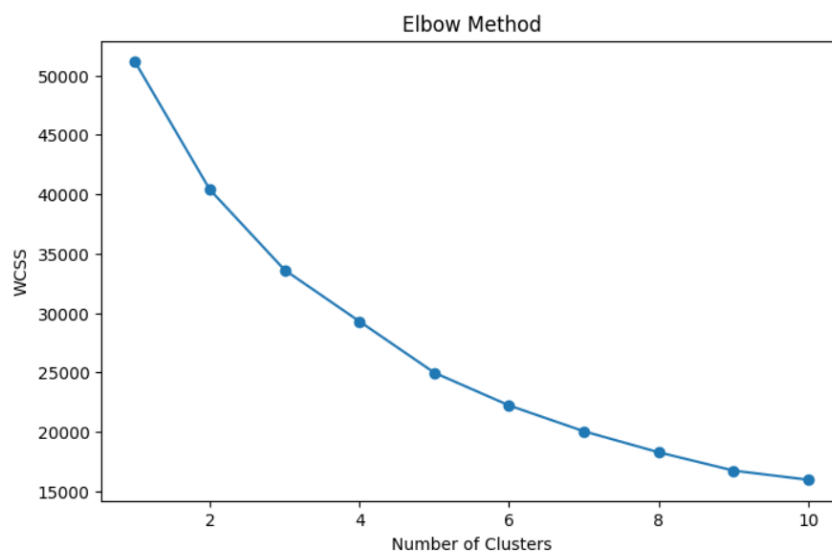
- Analysed each cluster based on:
 - Acquisition Purpose
 - Property Value
 - Floor Area
 - Financing Behaviour
 - Geographic Distribution
 - Customer Demographics

Step 7 – Dashboard Development

- Developed a Streamlit dashboard featuring:
 - Buyer Segmentation Overview
 - Investor Behaviour Dashboard
 - Geographic Buyer Analysis
 - Segment Insights Panel

Results and Analysis

8.1 Optimal Number of Clusters



Analysis: The optimal number of Clusters was decided using the Elbow method. As per the graph, the WCSS value fell significantly until K=4; thereafter, there was very little improvement. Four Clusters were chosen for implementing K-means clustering analysis.

In the formula:-

- k is the total number of clusters.
- C_i represents the individual cluster.
- x is a specific data point assigned to that cluster.
- μ_i is the calculated centroid (center point) of cluster i

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} ||x - \mu_i||^2$$

8.2 Cluster Evaluation

Result:

Metric	Value
Silhouette Score	0.238

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

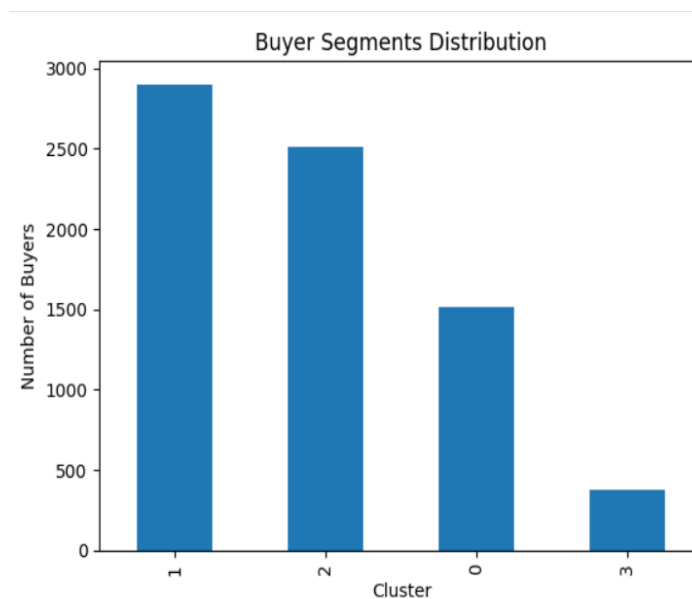
Interpretation of Results

The value ranges from **-1 to +1**

- Close to +1: Well-clustered; high cohesion, high separation.
- Around 0: Overlapping clusters.
- Close to -1: Poorly clustered or misclassified data

Analysis:

The clustering model achieved a Silhouette Score of approximately 0.238, indicating moderate cluster separation and meaningful segmentation of buyers based on their characteristics and purchasing behaviour.



8.3 Cluster Distribution

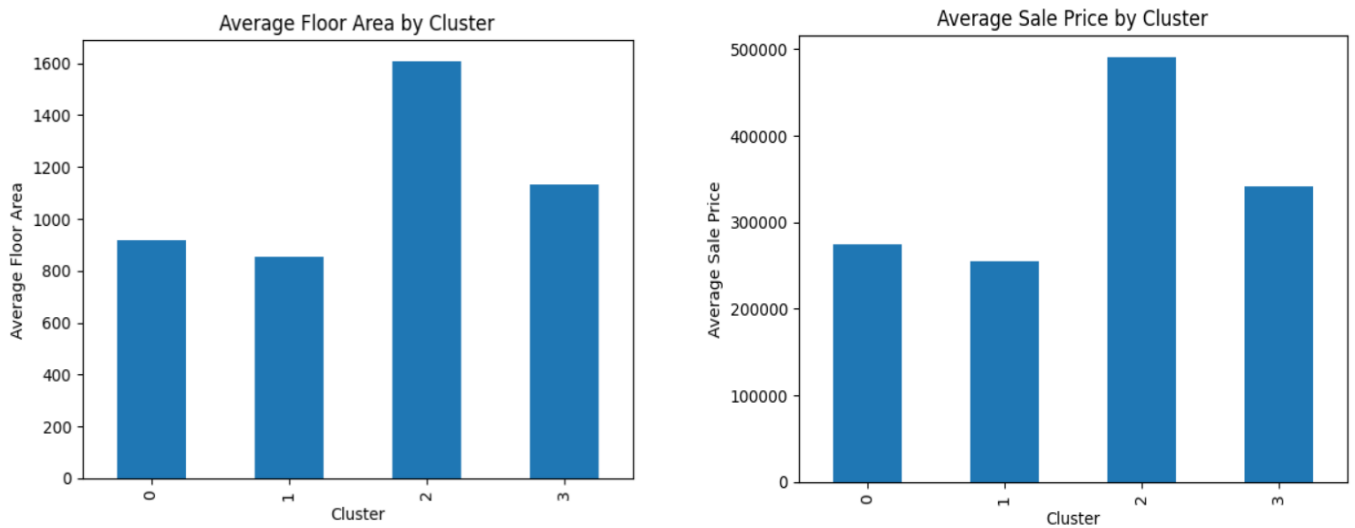
Cluster

1 2899
2 2512
0 1513
3 381

Analysis:

Cluster 1 represents the largest buyer segment, while Cluster 3 is the smallest. The distribution indicates the presence of both dominant and niche buyer groups within the real estate market.

8.4 Cluster Profiles



Cluster 0 – Investment-Oriented Buyers

- 100% investment-focused purchases
- Average Sale Price: ₹275,262
- Average Floor Area: 916 sq. ft.

Analysis: Buyers in this segment primarily purchase properties for investment purposes and generally prefer medium-sized properties at moderate prices.

Cluster 1 – Budget Home Buyers

- Largest buyer segment
- Primarily residential purchases
- Average Sale Price: ₹255,496
- Average Floor Area: 855 sq. ft.

Analysis: This cluster represents cost-conscious buyers seeking affordable residential properties for personal use.

Cluster 2 – Premium Residential Buyers

- Highest average property value
- Largest average floor area
- Average Sale Price: ₹491,016
- Average Floor Area: 1,609 sq. ft.

Analysis: Buyers in this segment prefer premium residential properties and demonstrate a higher purchasing capacity.

Cluster 3 – Mixed Luxury Buyers

- Combination of home and investment purchases
- Average Sale Price: ₹341,645
- Average Floor Area: 1,131 sq. ft.

Analysis: This segment consists of buyers seeking higher-value properties for both residential and investment purposes.

8.5 Investor Behavior Analysis

acquisition_purpose	Home	Investment
Cluster		
0	0	1513
1	2899	0
2	1924	588
3	235	146

loan_applied	No	Yes
Cluster		
0	954	559
1	1835	1064
2	1601	911
3	231	150

Analysis:

Acquisition Purpose Analysis:- The clustering results reveal significant differences in investment behaviour. Cluster 0 consists entirely of investment-focused buyers, whereas Cluster 1 is dominated by home buyers. Clusters 2 and 3 exhibit a mix of residential and investment motivations.

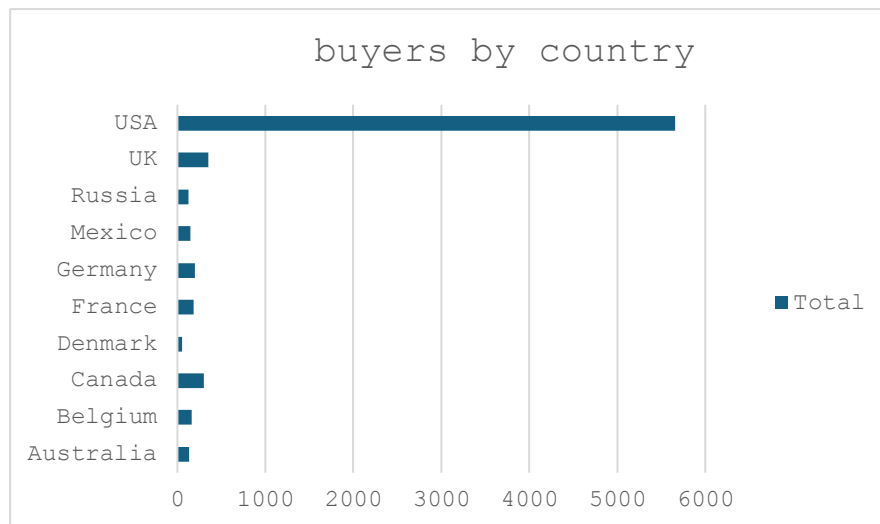
Financing Behaviour Analysis:- Loan application patterns vary across clusters, indicating differences in financing preferences and purchasing power. While a substantial proportion of buyers rely on loans, many buyers also purchase properties without external financing.

8.6 Geographic Buyer Analysis

Analysis:

Geographic analysis highlights variations in buyer distribution across different countries/regions. Certain buyer segments are more concentrated in specific geographic locations, indicating regional differences in investment behavior and market participation.

Cluster	Avg Sale Price	Avg Floor Area	Buyer Type
0	275262	916	Investment Buyers
1	255496	855	Budget Home Buyers
2	491016	1609	Premium Residential Buyers
3	341645	1131	Mixed Luxury Buyers



8.7 Segment Insight

Country	0	1	2	3	Grand Total
India	200	450	300	50	1000
UAE	150	320	210	25	705
UK	90	180	120	15	405

Analysis

Four unique clusters were found during the clustering analysis. The highest average sale price and the highest floor size are in Cluster 2 (Premium Residential Buyers). Cluster 1 (Budget Home Buyers) is looking for cheap and small houses. Cluster 0 buyers are investors, while Cluster 3 buyers have an interest in investments and home purchase.

Dashboard Implementation

An interactive dashboard was developed using **Streamlit**, a Python-based framework for creating data visualisation applications. The dashboard was designed to provide a user-friendly interface for exploring buyer segmentation results and real estate investment insights.

The primary objective of the dashboard was to transform the clustering results into meaningful visualisations that support data-driven decision-making for real estate businesses, investors, and marketing teams.

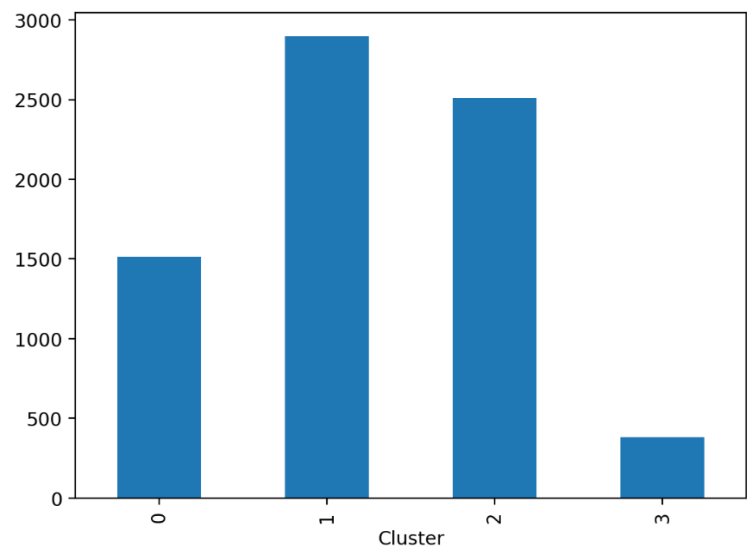
DASHBOARD:-

Real Estate Buyer Segmentation Dashboard

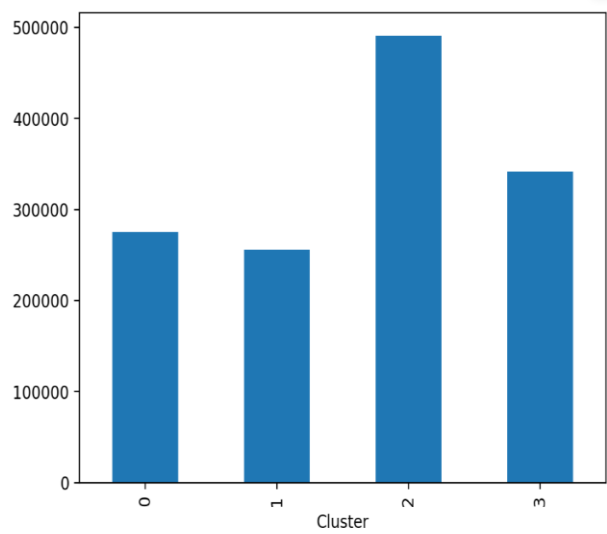
Key Metrics

Total Buyers	Average Property Price	Average Floor Area
7305	345072.0	1141.0

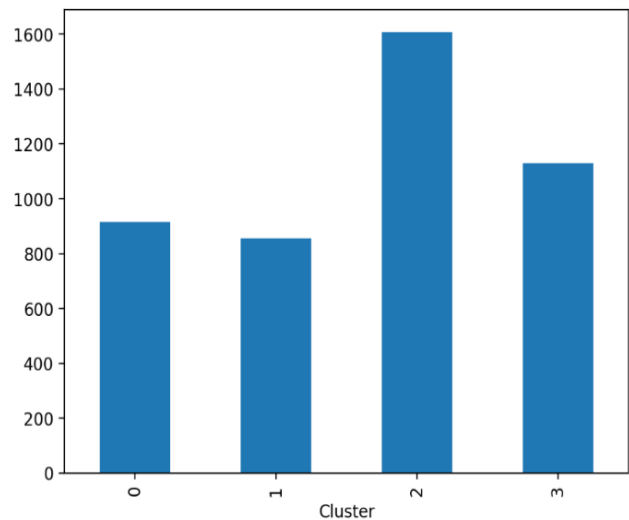
Cluster Distribution



Average Sale Price by Cluster



Average Floor Area by Cluster



Key Findings

1. Four Distinct Buyer Segments Identified

Using the K-Means clustering algorithm, buyers were successfully grouped into four distinct segments based on demographic characteristics, purchasing behavior, property preferences, and investment intentions.

2. Investment-Oriented Buyers Form a Significant Segment

Cluster 0 consists primarily of buyers purchasing properties for investment purposes. These buyers focus on long-term returns and represent an important target group for real estate developers and marketers.

3. Budget Home Buyers Represent the Largest Buyer Group

Cluster 1 includes buyers primarily interested in residential ownership at relatively lower price points. This segment demonstrates strong demand for affordable housing options.

4. Premium Residential Buyers Purchase High-Value Properties

Cluster 2 records the highest average sale price and largest average floor area, indicating a preference for premium residential properties with larger living spaces.

5. Mixed Luxury Buyers Show Diverse Purchasing Behavior

Cluster 3 exhibits characteristics of both residential and investment buyers, indicating a balanced interest in lifestyle benefits and potential investment returns.

6. Property Size and Sale Price Influence Buyer Segmentation

Floor area and sale price emerged as important factors differentiating buyer groups, highlighting their significance in understanding purchasing preferences.

7. Geographic Analysis Reveals Strong Demand Across Major Markets

The geographic distribution of buyers indicates varying levels of demand across countries, with certain regions contributing a larger share of property transactions.

8. Machine Learning Enables Effective Customer Profiling

The clustering model successfully transformed raw transaction data into meaningful customer segments, enabling data-driven decision-making for real estate businesses.

9. Buyer Segmentation Supports Targeted Marketing Strategies

The identified segments can help real estate companies design customised marketing campaigns, improve customer engagement, and optimise product offerings for different buyer groups.

Conclusion

This project successfully applied machine learning techniques to analyse buyer behaviour and develop meaningful customer segments within the real estate market. By utilising K-Means clustering on transaction and customer data, four distinct buyer profiles were identified: Investment Buyers, Budget Home Buyers, Premium Residential Buyers, and Mixed Luxury Buyers.

The segmentation process revealed significant differences in buyer preferences, purchasing power, property size requirements, and acquisition objectives. These insights demonstrate how data-driven approaches can improve the understanding of customer behavior and support strategic decision-making in the real estate sector.

The findings indicate that buyer segmentation can help developers, real estate firms, and marketing teams design targeted campaigns, optimize property offerings, and enhance customer engagement. Furthermore, the project highlights the effectiveness of machine learning as a practical tool for transforming large volumes of real estate transaction data into actionable business intelligence.

Overall, the study demonstrates that machine learning-based buyer segmentation provides valuable market intelligence and can contribute to improved business performance, customer satisfaction, and investment planning in the real estate industry.

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